# Stock Status and Management Challenges of Bigeye Tuna (*Thunnus obesus*) in the Eastern Pacific Ocean

Fox Surplus Production Model · CPUE Analysis · MSY Reference Points · IATTC Data (1960–2023)

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Tools: R (tidyverse, ggplot2) · IATTC Public Database
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| 56,300 t | 11,876 | 81% |
|---|---|---|
| Maximum Sustainable Yield | Sets — current effort is 28% above | Y/MSY ratio (2019–2023 mean) |
| Fox model MSY estimate | Effort at MSY (E_MSY) | Recent catch vs MSY |


## Abstract

Bigeye tuna (*Thunnus obesus*) is one of the most valuable tropical tunas in the Eastern Pacific Ocean (EPO). This report evaluates its stock status and management challenges using stock status indicators (SSIs), surplus production modeling, and harvest strategy evaluations. The analysis focuses on data from 1990 onward, with effort and catch trends estimated from available fishery-dependent records from the IATTC public database (1960–2023).

Findings indicate that the bigeye tuna stock is not currently overfished, and fishing mortality is close to but still below the established reference limits. However, fishing effort remains excessive, particularly due to the high impact of floating-object sets that generate considerable juvenile mortality. The stock is thus maintained by a narrow margin of sustainability, where even modest increases in fishing pressure or declines in recruitment could compromise long-term viability.

The report highlights the urgent need to reduce fishing capacity to more efficient levels, strengthen harvest strategies, and minimize juvenile mortality. It also stresses the importance of preparing for climate-related uncertainties that could alter productivity and habitat distribution in the coming decades.


## 1. Introduction

The genus *Thunnus*, belonging to the family Scombridae, comprises eight species commonly known as tunas (Collette et al., 2001). Several of these species are widely traded internationally, including the Atlantic bluefin (*T. thynnus*), Pacific bluefin (*T. orientalis*), Southern bluefin (*T. maccoyii*), bigeye (*T. obesus*), yellowfin (*T. albacares*), and albacore (*T. alalunga*). Despite being among the most important species for the global fishing industry, tunas are also among the most threatened fish species worldwide (Viñas & Tudela, 2009).

Bigeye tuna (*T. obesus*) is a circumglobal species recorded in both tropical and temperate seas, absent from the Mediterranean Sea. It is a pelagic and oceanodromous species found in waters ranging from 13°C to 29°C, with an optimum temperature range between 17°C and 22°C. Its distribution extends as far north as 55–60° in the boreal summer and as far south as 45–50° in the austral summer, though juveniles and reproductively active adults are primarily found in tropical waters (Powell et al., 2009; Collette & Graves, 2019). During the night, bigeye tuna typically inhabits waters above the mixed layer, while during the day they often undertake prolonged dives to depths of 250–500 m to target deep scattering layer prey (Dagorn et al., 2000).

This species is a multiple spawner, capable of spawning every one or two days over several months (Nikaido et al., 1992). Spawning events are associated with the full moon and occur year-round in the western Pacific (Sun et al., 2013). Although spawning appears to occur widely across the equatorial Pacific, the highest reproductive potential is believed to occur in the eastern Pacific, based on maturation patterns, size frequency data, and catch per unit of effort (Kikawa, 1966).

Bigeye tuna is an important fishery resource managed across the Atlantic, Indian, Western and Central Pacific, and Eastern Pacific Oceans. However, due to insufficient management measures, overfishing is occurring in approximately 40% of its global range, with estimated declines of at least 20% in the Western and Central Pacific and Eastern Pacific, and up to 40% in Atlantic and Indian Ocean stocks over the past two decades. As a result, the species is listed as Vulnerable on the IUCN Red List (Collette et al., 2021).

In the Eastern Pacific Ocean, the stock represents approximately 22% of the global population (ISSF, 2020). Purse seines currently account for approximately 74% of landings and longlines for 26%. Increased selectivity for smaller fish has reduced the estimated MSY by almost 50% (Collette & Graves, 2019; IATTC, 2020). Given these concerns, this report aims to evaluate the catch–effort relationship for bigeye tuna in the EPO, fit the Fox surplus production model to estimate $E_{MSY}$ and MSY, and assess the relative exploitation status of the stock.

## 2. Objectives

- Analyze temporal trends in catch, fishing effort, and CPUE for bigeye tuna in the EPO (1960–2023).
- Disaggregate CPUE by set type (OBJ, NOA, DEL) to identify patterns in stock productivity.
- Fit Schaefer and Fox surplus production models to OBJ set data (1990–2023) and evaluate model performance.
- Estimate MSY and $E_{MSY}$ reference points from the best-fitting model.
- Compare current fishing effort and catch against MSY benchmarks to assess exploitation status.

## 3. Methods

### 3.1 Data Source

Catch and effort data for bigeye tuna (*Thunnus obesus*) in the EPO were obtained from the Inter-American Tropical Tuna Commission (IATTC) public database. The dataset comprised annual catch (metric tons) and fishing effort (number of sets) from purse seine vessels operating with floating-object sets (OBJ), unassociated sets (NOA), and dolphin sets (DEL) between 1960 and 2023.

### 3.2 Data Processing

Raw data were filtered to include only OBJ set type records for surplus production modeling, to ensure homogeneity in fishing mode and selectivity patterns. Annual CPUE was calculated as total annual catch (mt) divided by total annual effort (sets). Effort was expressed in standardized units (number of sets) to allow direct comparison across years.

### 3.3 Model Selection and Fitting

To assess the catch–effort relationship, both the Schaefer and Fox surplus production models were fitted to the 1990–2023 OBJ data. The Fox model (Fox, 1970) assumes a logarithmic relationship between CPUE and effort: $\ln(CPUE) = a + b \times E$, where a is the intercept related to stock intrinsic productivity and b is the slope describing the effect of effort on CPUE. Parameters were estimated using ordinary least squares (OLS) regression. The analysis period was restricted to 1990 onward to ensure consistency, as earlier records showed limited coverage and higher uncertainty.

### 3.4 Reference Point Estimation

From the fitted Fox model, $E_{MSY} = -1/b$ and MSY were estimated. Relative exploitation status was assessed by comparing the most recent five-year average fishing effort ($E_{current}$) to $E_{MSY}$, and the most recent yield ($Y_{current}$) to MSY. All analyses and visualizations were performed in R version 4.5.0 using packages *tidyverse* and *ggplot2*.

| Component | Source / Tool | Details |
| --- | --- | --- |
| Catch and effort data | IATTC public database | 1960–2023 · Purse seine · OBJ/NOA/DEL |
| CPUE calculation | R (tidyverse) | Catch (mt) / Effort (sets) per year |
| Schaefer model | OLS regression (R) | Fit period: 1990–2023 · OBJ only |
| Fox model | OLS regression (R) | ln(CPUE) = a + b·E · Fit period 1990+ |
| Reference points | Fox model estimates | MSY and E_MSY from fitted parameters |
| Visualization | R (ggplot2) | Time series, yield curves, status plots |

*Table 1. Summary of data sources and analytical methods used in this study.*

# 4. Results

## 4.1 Catch Trends

Total catches of bigeye tuna (*Thunnus obesus*) in the EPO increased steadily during the 1990s, rising from approximately 45,000 t in 1990 to a maximum of about 80,000 t in 2000 (Figure 1). After this peak, catches fluctuated between 60,000 and 70,000 t until the mid-2010s, followed by a decline in the most recent years to around 35,000–55,000 t. This pattern indicates a rapid expansion phase, stabilization, and recent contraction of yields, consistent with a fishery operating near maximum capacity.

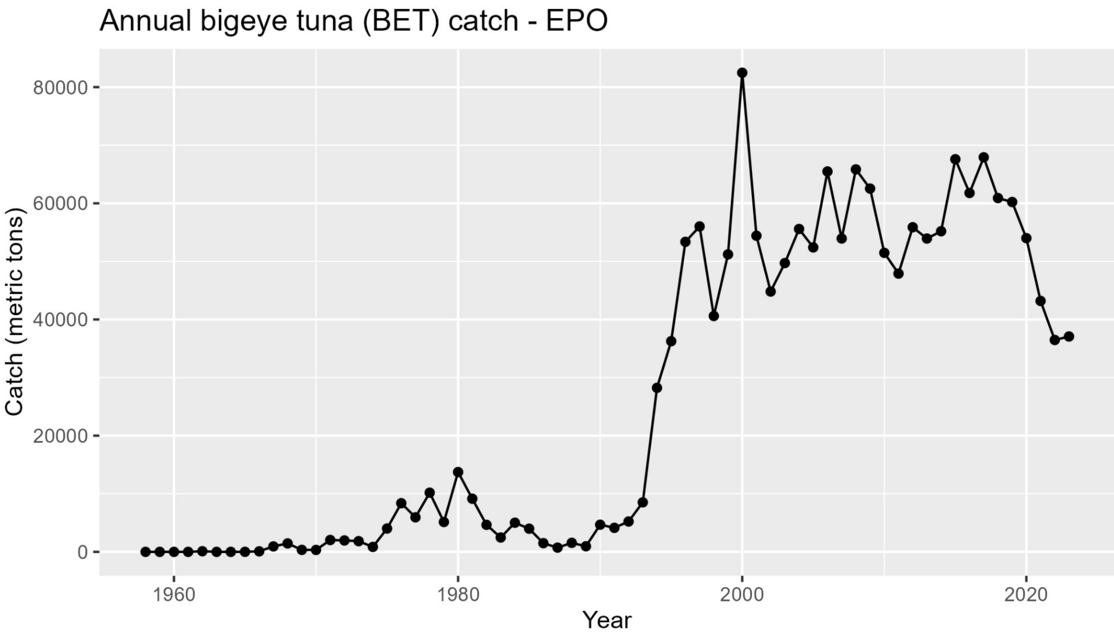



*Figure 1. Annual bigeye tuna (BET) catch (metric tons) in the EPO from 1955 to 2023. The sharp rise in the 1990s, peak in 2000, and recent decline are evident. Source: IATTC. Analysis: Israel Vallejo Muñoz.*

## 4.2 Effort Dynamics

Fishing effort, expressed as the number of purse-seine sets, showed a strong increasing trend throughout the study period (Figure 2). Effort grew from fewer than 10,000 sets in 1960 to over 30,000 sets by the early 2020s. The most pronounced increase occurred after 2000, largely due to the expansion of floating-object sets (OBJ). While catches stabilized and then declined, effort continued to grow, evidencing declining productivity of the stock.

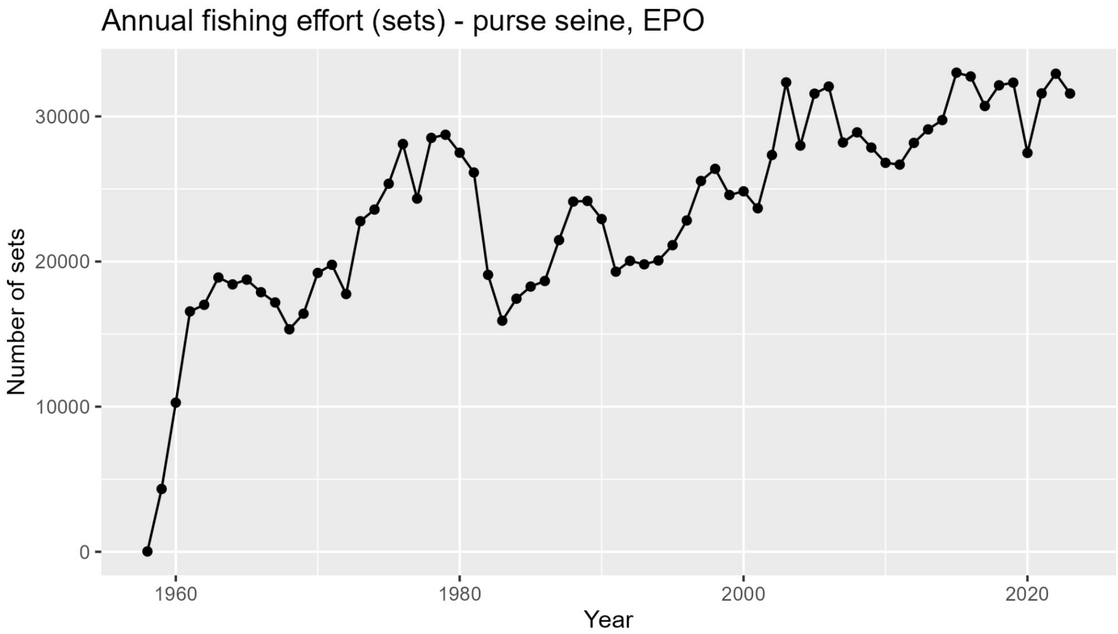



*Figure 2. Annual fishing effort (number of purse-seine sets) targeting bigeye tuna in the EPO from 1955 to 2023. Source: IATTC. Analysis: Israel Vallejo Muñoz.*

### 4.3 Catch per Unit Effort (CPUE)

The annual CPUE of bigeye tuna in the EPO (Figure 3) exhibited considerable variability throughout the study period. Values were low during the 1980s and early 1990s, before increasing sharply. The maximum CPUE was observed in 2000, reaching 3.32 t/set, after which a general decline became evident. In recent years, CPUE has stabilized at comparatively lower levels, indicating reduced fishing efficiency and potential signs of increased exploitation pressure.

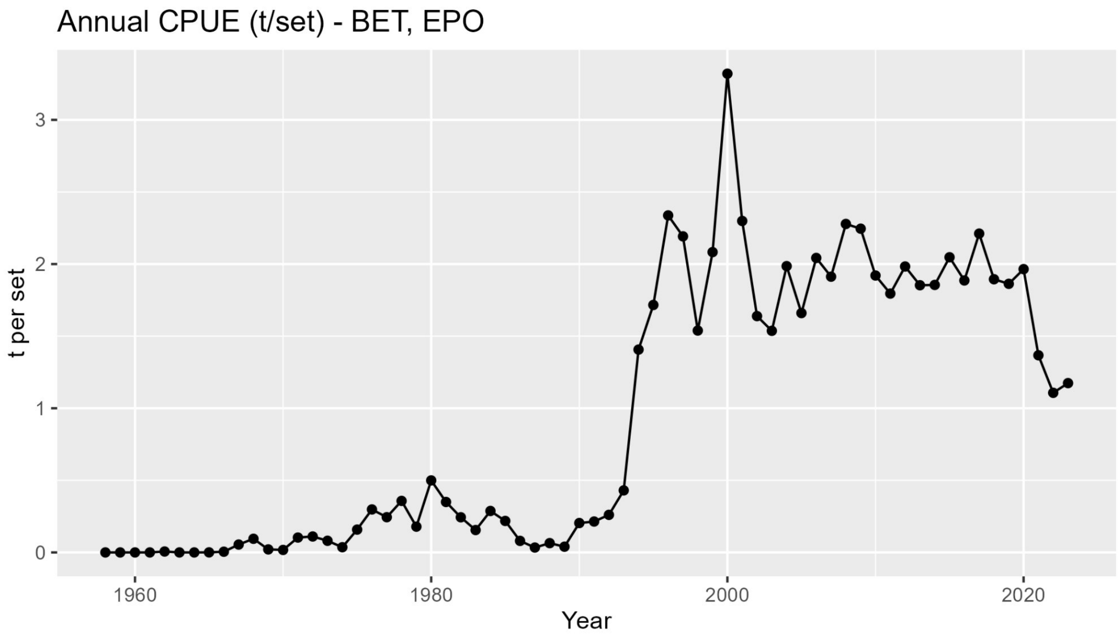



*Figure 3. Annual CPUE (t/set) of bigeye tuna in the EPO from 1955 to 2023. The peak in 2000 and subsequent decline are consistent with increasing fishing pressure. Source: IATTC. Analysis: Israel Vallejo Muñoz.*

### 4.4 CPUE by Set Type

When disaggregated by set type (Figure 4), a clear contrast emerges among the three purse-seine modes. OBJ sets peaked at 18.8 t/set in 2000 but declined steeply to only 2.16 t/set by 2023. NOA sets maintained consistently low values (0.04–0.08 t/set in recent years), contributing marginally to the overall catch. DEL sets showed negligible values throughout the time series, with a historical maximum of 0.079 t/set in 1986 and near-zero values (<0.005 t/set) in the last decade. This contrast highlights that the heavy reliance on OBJ sets has largely driven and amplified the observed decline in stock productivity.

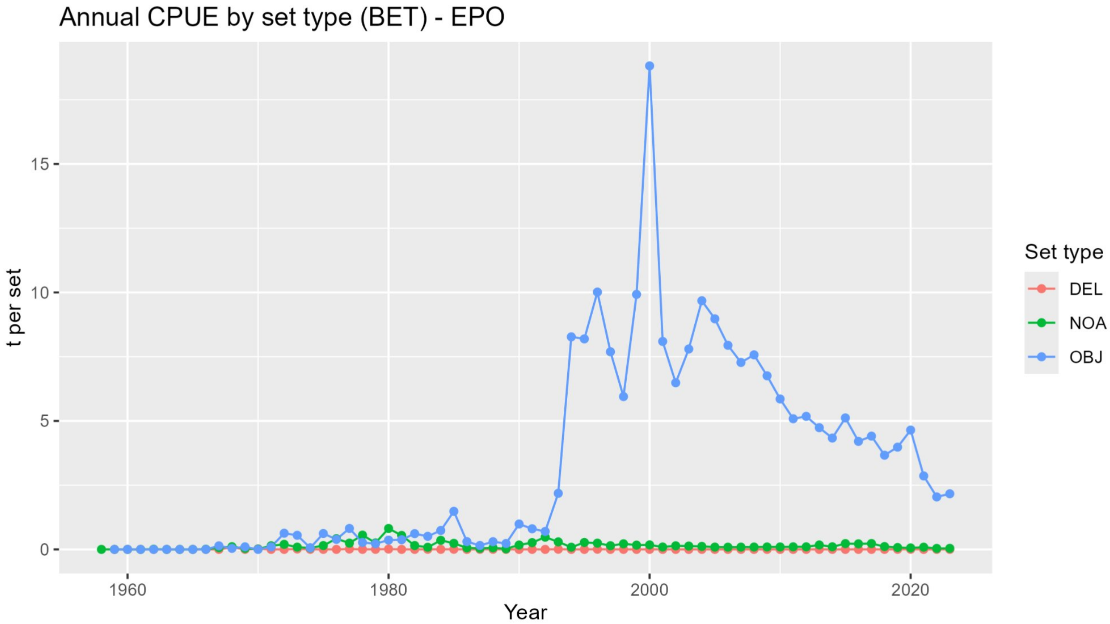


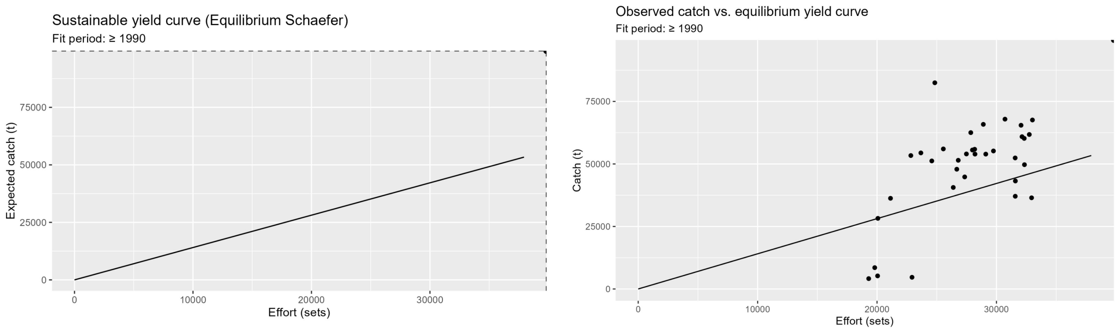

*Figure 4. Annual CPUE (t/set) of bigeye tuna in the EPO by set type (OBJ = floating-object associated, NOA = unassociated, DEL = dolphin-associated) from 1955 to 2023. Source: IATTC. Analysis: Israel Vallejo Muñoz.*

## 4.5 Surplus Production Models

The analysis was restricted to the period from 1990 onward to ensure consistency and reliability in the data series, and only OBJ set data were used, as OBJ sets have historically represented the predominant fishing strategy for bigeye tuna in the EPO.

### 4.5.1 Schaefer Model

The Schaefer model produced a poor fit to the catch–effort data (Figures 5–6). The relationship was almost linear with an $R^2$ close to 0.1, failing to capture the expected dome-shaped yield curve. This indicates that the symmetric assumptions of the Schaefer formulation do not adequately represent the stock dynamics of bigeye tuna in this region.




*Figures 5–6. Left: Equilibrium Schaefer yield curve (fit period 1990+). Right: Observed catch vs. effort with Schaefer model fit. The near-linear relationship confirms poor model fit ($R^2 \approx 0.1$). Source: IATTC. Analysis: Israel Vallejo Muñoz.*

### 4.5.2 Fox Model Results

In contrast, the Fox model provided a substantially stronger fit, with an $R^2$ of 0.50 (Figure 7). The equilibrium yield curve displayed the expected dome-shaped form, consistent with theoretical assumptions for a species near maximum exploitation. Model estimates were as follows:

- Maximum Sustainable Yield (MSY): **56,300 t**
- Effort at MSY ($E_{MSY}$): **11,876 sets**

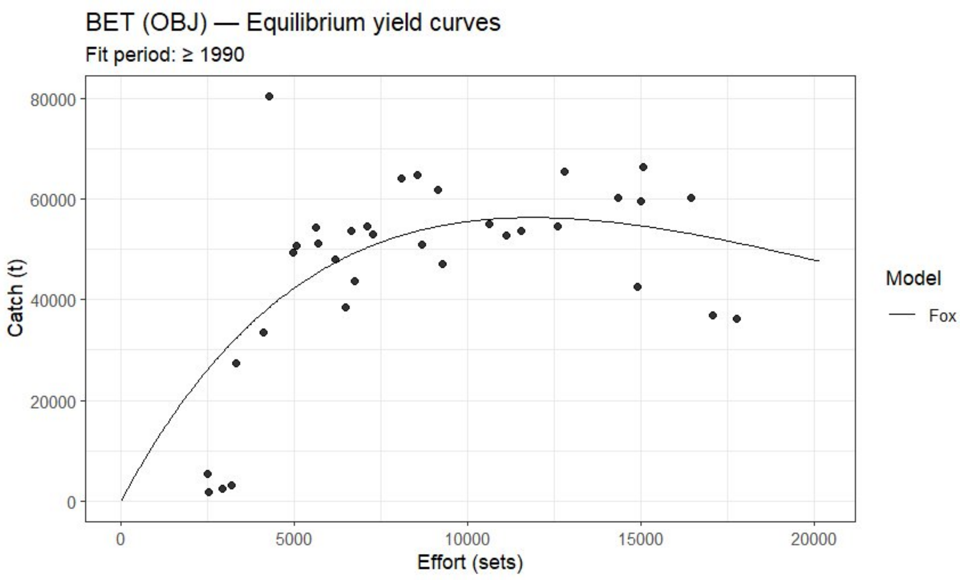



*Figure 7. Fox equilibrium yield curve for BET (OBJ sets, fit period 1990+). The dome-shaped curve confirms a better fit than the Schaefer model ($R^2 = 0.50$). Source: IATTC. Analysis: Israel Vallejo Muñoz.*

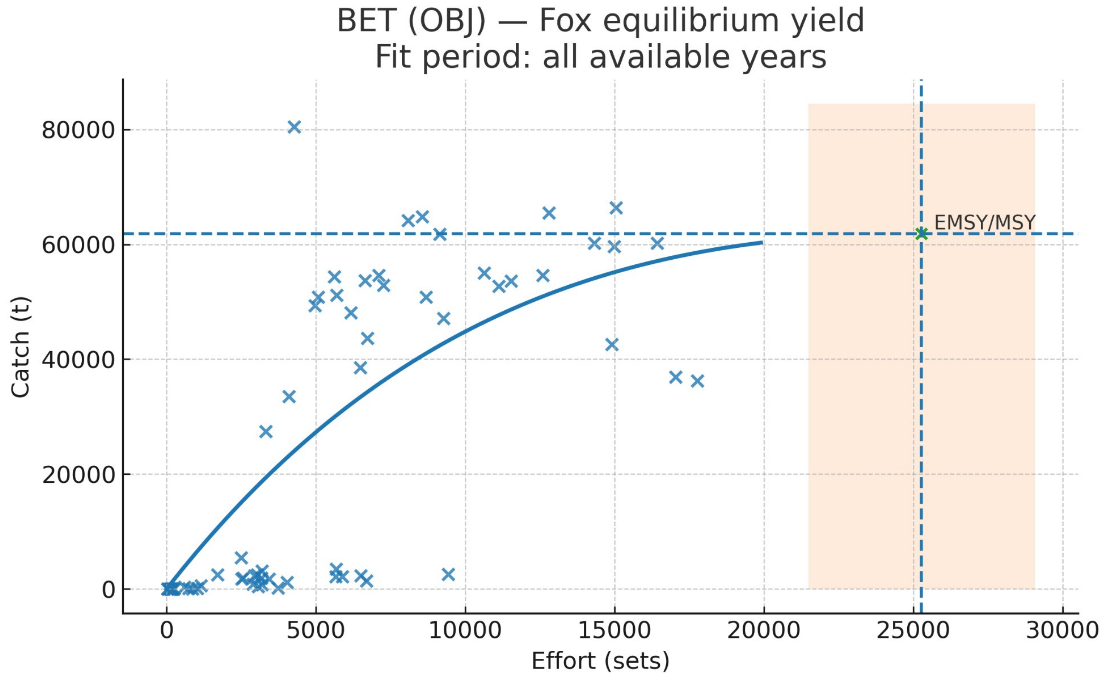



*Figure 8. BET (OBJ) Fox equilibrium yield curve — all available years. The EMSY/MSY reference point is marked. The shaded zone indicates the range of recent effort levels relative to optimal. Source: IATTC. Analysis: Israel Vallejo Muñoz.*

### 4.6 Status Relative to Reference Points

During the most recent period (2019–2023), average fishing effort was 15,249 sets, approximately 28% above $E_{MSY}$. The mean annual catch was 45,811 t, representing 81% of MSY. This pattern indicates that the stock is subject to overexploitation in terms of effort ($E/E_{MSY}$ = 1.28) but is underperforming in yield (Y/MSY = 0.81). While catches remain close to the theoretical maximum, they are being achieved at a disproportionately high level of fishing effort, indicating declining stock productivity.

| Indicator | Recent value (2019–2023) | Reference point | Ratio |
|---|---|---|---|
| Mean annual catch | 45,811 t | MSY = 56,300 t | Y/MSY = 0.81 |
| Mean annual effort | 15,249 sets | E_MSY = 11,876 sets | E/E_MSY = 1.28 |
| OBJ CPUE (2023) | 2.16 t/set | Peak OBJ = 18.8 t/set | ~11.5% of peak |

*Table 2. Summary of recent stock status indicators relative to MSY-based reference points.*

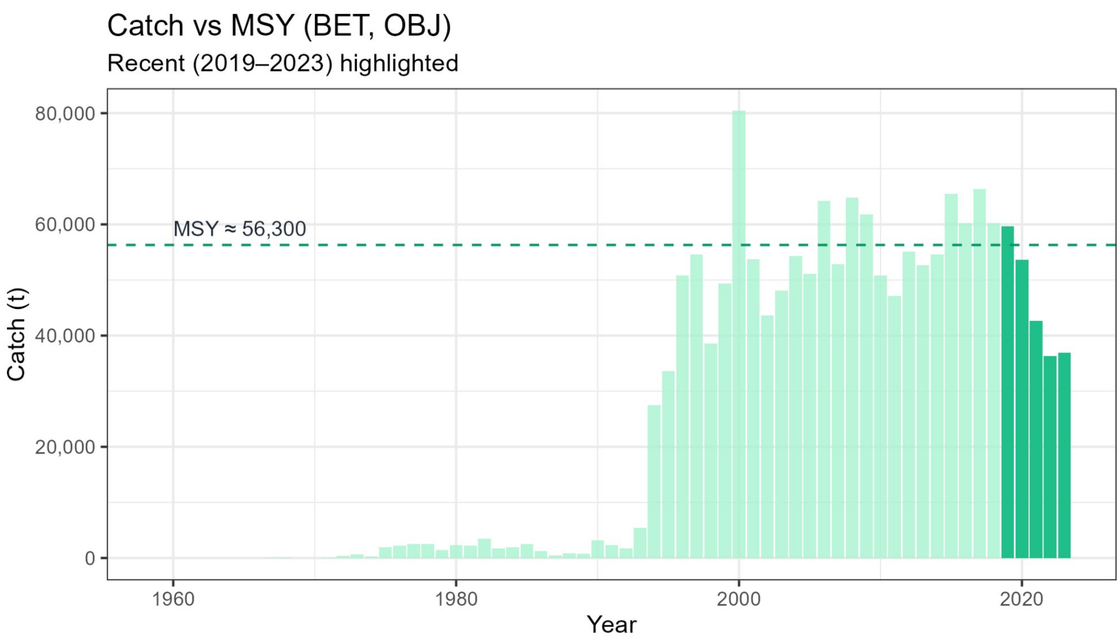



*Figure 9. Annual catch vs. MSY reference (dashed line ≈ 56,300 t) for BET (OBJ sets). Recent years 2019–2023 highlighted in dark green. Source: IATTC. Analysis: Israel Vallejo Muñoz.*

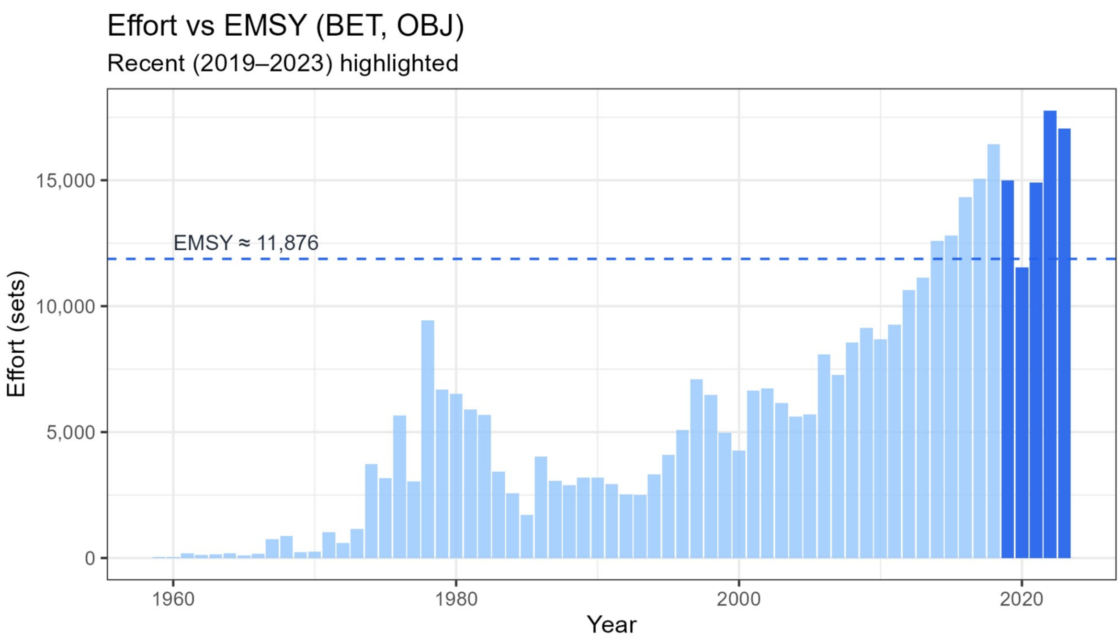



*Figure 10. Annual OBJ fishing effort vs. E_MSY reference (dashed line ≈ 11,876 sets). Recent years (2019–2023) highlighted in dark blue, consistently exceeding E_MSY. Source: IATTC. Analysis: Israel Vallejo Muñoz.*

## 5. Discussion

The present evaluation of bigeye tuna (*Thunnus obesus*) in the EPO provides a nuanced view of a stock that, although currently assessed as sustainable, is subject to high levels of fishing pressure and ecological vulnerability. Bigeye tuna is one of the three main tropical tunas exploited in the EPO, alongside yellowfin and skipjack. Unlike skipjack, which is characterized by fast growth and high resilience to fishing, bigeye exhibits slower growth, later maturity, and higher market value, making it particularly sensitive to overexploitation (ISSF, 2022). For this reason, both ISSF and the IATTC have consistently highlighted bigeye as the species of greatest concern among tropical tunas in the region (IATTC, 2022a; IATTC, 2023a).

**Key Finding: The Fox model estimates MSY at 56,300 t with an optimal effort of 11,876 sets. However, observed fishing effort in recent years is around 15,000–18,000 OBJ sets (and ~30,000 total sets), far exceeding E_MSY. While catch (~45,800 t mean) remains below MSY, it is being sustained at the cost of disproportionate effort, indicating declining stock productivity and a narrow margin of sustainability.**

These findings are consistent with the most recent official benchmark assessment by the IATTC (IATTC, 2024a), which concluded that the stock is not overfished and not subject to overfishing, with reference point indicators $S/S_{MSY} \approx 1.16$ and $F/F_{MSY} \approx 0.94$ for 2022. This indicates that spawning biomass remains above the critical threshold, while fishing mortality is just below $F_{MSY}$. However, the proximity of these indicators to their limit values leaves little margin for error, and any increase in effort or environmental stress could rapidly push the stock into an overexploited condition.

A major concern highlighted in both this analysis and SAC reports is the disproportionate impact of floating-object purse-seine sets (OBJ) on juvenile bigeye. While skipjack is the primary target of OBJ sets, significant quantities of juvenile bigeye are caught incidentally, contributing heavily to overall fishing mortality (IATTC, 2022b; IATTC, 2022c). This bycatch of juveniles not only reduces recruitment to the spawning stock but also exacerbates the inefficiency of the fishery, as much of the potential yield is harvested before fish reach marketable sizes.

Historical trends reveal a long-term increase in fishing effort since the 1990s, driven by the expansion of purse-seine fleets and technological advances in fish aggregating devices (FADs) (ISSF, 2022; IATTC, 2023b). Although catches have stabilized near the MSY level, effort has continued to grow, reinforcing concerns about long-term sustainability. The IATTC is implementing a harvest strategy based on harvest control rules (HCRs) and management strategy evaluation (MSE) to ensure that fishing mortality remains within precautionary bounds (IATTC, 2024b).

Environmental variability further complicates the outlook. Ocean warming, deoxygenation, and changes in prey distribution may alter habitat suitability and migratory patterns, potentially affecting stock productivity and the spatial overlap with fishing fleets. These changes could amplify the risk of recruitment failure or regional depletion, particularly if management measures are not sufficiently precautionary.

## 6. Conclusions

The present analysis confirms that bigeye tuna (*Thunnus obesus*) in the Eastern Pacific Ocean is currently in a sustainable state, with biomass above the threshold that would indicate overfishing and fishing mortality slightly below critical limits. However, this apparent stability masks significant vulnerabilities:

- The fishery is characterized by excessive capacity: fishing effort far exceeds the level needed to achieve MSY efficiently.
- High juvenile mortality associated with OBJ sets reduces recruitment and long-term yield potential.
- Current stock indicators are close to their limit reference points, leaving a narrow buffer of sustainability.
- Growing climate uncertainty may alter habitat suitability and productivity in ways that are not yet fully captured by current models.

Ensuring sustainability will therefore require decisive action: reducing fishing effort to more efficient levels, minimizing juvenile mortality, and adopting adaptive harvest strategies based on precautionary reference points. These measures would not only safeguard the ecological sustainability of the stock but also enhance the long-term economic viability of the fishery, ensuring that bigeye tuna in the Eastern Pacific Ocean remains a valuable resource for both current and future generations.

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